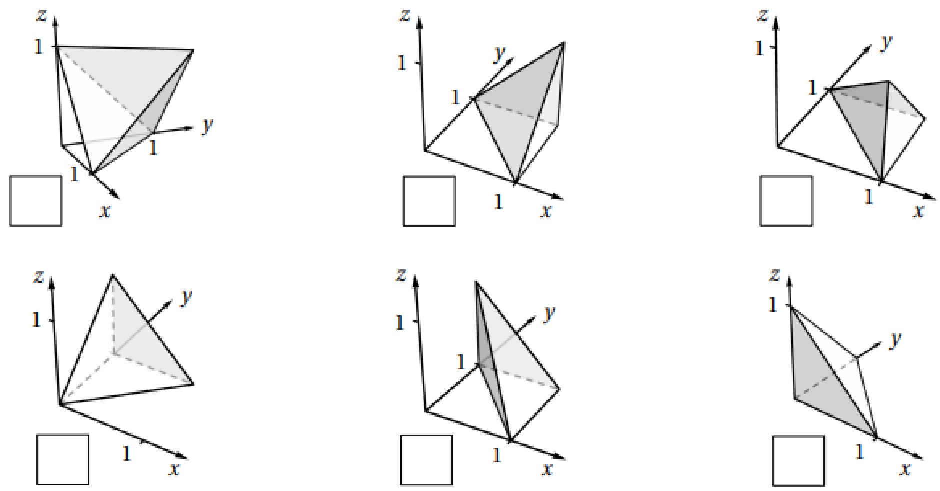
MTH 202 Chapter 4: Multiple Integrals

4.5 Triple Integrals Worksheet

1. The integral of the function $f(x,y,z)=2x$ over a region R is computed by $\int_0^1\int_0^y\int_0^{y-x} 2x\,dz\,dx\,dy$.

a) Mark the box below of the picture of the region R.



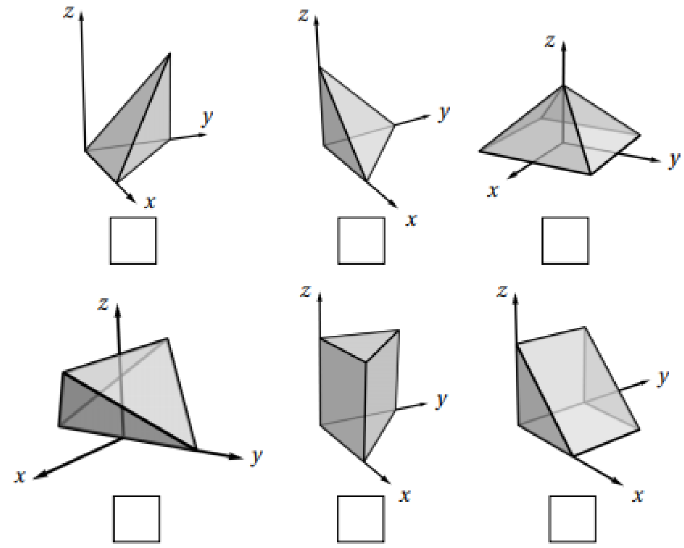
b) Evaluate the iterated integral from part (a).

c) Re-express this iterated integral with differentials in the order of *dzdydx*.

d) Re-express this iterated integral with differentials in the order of *dydzdx*.

2. The integral of the function $f(x,y,z) = xyz$ over a region R is computed by $\int_0^1 \int_0^{1-x} \int_0^1 f(x,y,z)\,dz\,dy\,dx$.

a) Mark the box below of the picture of the region R.



b) Evaluate the iterated integral from part (a).

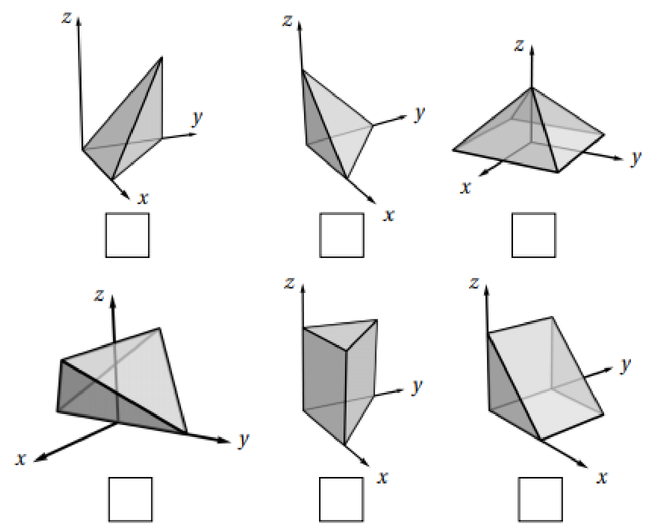
c) Re-express this iterated integral with differentials in the order of *dzdxdy*.

d) Re-express this iterated integral with differentials in the order of *dxdydz*.

3. The integral of the function $g(x,y,z) = xyz\sqrt{1+x^2}$ over a region R is computed by

$$\int_0^1 \int_{-1+z}^{1-z} \int_{-z}^{z} g(x,y,z)\,dx\,dy\,dz.$$

a) Mark the box below of the picture of the region R.



b) Evaluate the iterated integral from part (a).

c) Explain why this iterated integral cannot be re-expressed as a single integral with differentials in the order of *dxdzdy*. Then, re-express this iterated integral as the sum of two integrals with differentials in the order of *dxdzdy*.

d) Re-express this iterated integral as the sum of two iterated integrals with differentials in the order of *dzdydx*.

4. Consider the solid determine by the region $E=\left\{(x,y,z)\,|\,y^2\le x\le 2,\ 0\le y\le\sqrt{2},\ 0\le z\le 1-\frac{1}{2}x+\frac{1}{\sqrt{2}}y\right\}$

a) Draw a picture of the solid communicated by this integral.

b) Set up an iterated integral which represents the volume of this region.

c) Rewrite this iterated integral so that the differentials are in the order: $dzdydx$.

5. Suppose $f(x,y,z)$ is a continuous function. Consider the iterated integral:

$$\int_0^1 \int_0^{\cos\left(\frac{\pi}{2}x\right)} \int_0^{1-x-\frac{1}{2}y} f(x,y,z)\,dz\,dy\,dx \,.$$

a) Draw a picture of the solid communicated by this integral.

b) Rewrite this iterated integral so that the differentials are in the order: $dzdxdy$.

c) If E represents the region of integration, then determine the projection of E onto the xz-plane.

d) If E represents the region of integration, then determine the projection of E onto the yz-plane.

6. Suppose $g(x,y,z)$ is a continuous function. Consider the iterated integral: $\int_0^2 \int_0^{1-\frac{1}{2}x} \int_0^y g(x,y,z)\,dz\,dy\,dx$.

a) Draw a picture of the solid communicated by this integral.

b) Rewrite this iterated integral so that the differentials are in the order: $dzdxdy$.

c) Rewrite this iterated integral so that the differentials are in the order: $dydzdx$.

d) Rewrite this iterated integral so that the differentials are in the order: $dxdydz$.

7. Let R be the region in the first octant lying below the plane $x+y+z=1$.

(a) Setup a double integral which computes the volume of R with the differentials $dxdy$.

(b) Setup a triple integral which computes the volume of R with the differentials $dydxdz$.

8. Find the volume of the solid bounded by the coordinate planes and the plane $x+2y+3z=6$.

9. Suppose $f(x,y,z)$ is a continuous function. Consider the iterated integral: $\int_0^4 \int_{\sqrt{x}}^{5-\sqrt{x}} \int_0^{1-\frac{1}{4}x} f(x,y,z)\,dz\,dy\,dx$.

a) Draw a picture of the solid communicated by this integral.

b) Rewrite this iterated integral so that the differentials are in the order: $dydxdz$.

c) If E represents the region of integration, then determine the projection of E onto the yz-plane.